

PROJECT CENTRIC LEARNING (PCL) PHASE 2

			PROJECT TEAM NO.	CoEPCL-0205
PROJECT TITLE		Deepfake Detection and Blockchain Integration		
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DETAILS OF THE PROJECT GUIDE				
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DOMAIN OF THE PROJECT				
Machine Learning / Deep Learning and Blockchain Technology				
Scope of the Project				
1. Deepfake Detection: Classify uploaded media (images/videos) as REAL or FAKE using a ResNet-50 CNN model with transfer learning.				
2. Blockchain Verification: Generate SHA-256 hash of uploaded files and store on Ethereum testnet (Sepolia) via Solidity smart contracts for tamper-proof integrity verification.				
3. Full-Stack Web Application: Develop a Flask-based backend with REST APIs and an HTML/CSS/JavaScript frontend for user-friendly media upload, detection, and verification.				

PROJECT GUIDE

PROJECT COORDINATOR

Existing System:-

- Existing deepfake detection systems mainly use traditional image processing methods or basic CNN classifiers.
- These methods often analyze individual frames instead of extracting deeper pretrained visual features.
- As a result, they may fail to accurately detect newer and more realistic deepfakes generated using advanced GAN models.
- Most current systems focus only on detection and do not verify the long-term integrity of media files.
- There is no immutable record to prove whether a media file has been altered after its first authentication.
- This makes post-verification difficult and weakens trust in digital media.
- Existing deepfake detection solutions generally do not integrate blockchain or distributed ledger technology.
- Because of this, detection results remain centralized and may be vulnerable to manipulation, deletion, or loss.

Problem Identification:-

- The rapid growth of AI-generated synthetic media, especially deepfakes, threatens digital trust, journalism, cybersecurity, and personal identity.
- Deepfake images and videos are becoming so realistic that humans often cannot distinguish them from genuine media.
- There is a strong need for an automated and highly accurate system to detect deepfake images and videos.
- There is also a need for an immutable and decentralized verification mechanism to ensure authenticated media is not modified later.
- Without such a system, verified media can still be altered after authentication without reliable proof of tampering.
- Deepfakes can be misused for spreading misinformation and fake news.
- They can also enable identity fraud, impersonation, and reputational damage.
- In legal and investigative contexts, manipulated media may lead to evidence tampering and reduced trust in digital proof.

Proposed System:-

- The proposed system combines Machine Learning and Blockchain Technology for deepfake detection and media integrity verification.
- It uses a ResNet-50 CNN model with transfer learning for classifying media as REAL or FAKE.
- The model is pretrained on ImageNet and fine-tuned using deepfake datasets such as FaceForensics++ and CelebDF.
- The detection system achieves approximately 97% accuracy.
- For integrity verification, the system uses Ethereum blockchain, specifically the Sepolia testnet.
- Solidity smart contracts store the SHA-256 hash of uploaded files on the blockchain.
- This creates a tamper-proof and immutable record of the verified media.
- The backend is developed using a Flask-based REST API.
- The frontend is built with HTML, CSS, and JavaScript for simple user interaction.
- Users can upload images or videos for deepfake detection.
- The system provides real-time detection results along with confidence scores.
- Users can verify the authenticity of the same file at any later time using the immutable blockchain ledger

No. of Modules

Sl. No.	Name of the Module	Remarks
1	Frontend Module (UI)	Employs YOLOv11s-seg equivalent (ResNet-50): Achieves real-time media classification with high accuracy for REAL/FAKE detection.
2	ML Service Module (ResNet-50)	Transfer Learning Pipeline: Pretrained ImageNet weights fine-tuned on FaceForensics++ / CelebDF for domain-specific deepfake detection.
3	Blockchain Service Module	SHA-256 Hash Registry: Computes and stores file hashes on Ethereum Sepolia testnet via
4	Backend API s Storage Module (Flask)	REST API + Web UI: Flask endpoints (/upload, /verify, /health) with HTML/JS frontend delivering detection results and blockchain verification status.